

Performative Prediction in a Single-Asset Market

*Self-Fulfilment, Signal Erosion, and Economic
Value under Forecast Adoption*



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Abstract

This report studies how widespread adoption of a shared autoregressive forecasting rule affects forecast performance and economic value in a single-asset agent-based market. The market combines Brock-Hommes mean-variance demand with a Beja-Goldman / Farmer-Joshi market-maker price update. Traders may switch from a no-skill null rule to a public rolling AR forecast under either exogenous stochastic diffusion or endogenous certainty-equivalent switching. The forecast is evaluated against two distinct targets on the same path: the realised return, which is the metric a deployed forecaster actually sees, and a demand-adjusted counterfactual that strips out contemporaneous price impact.

The simulator identifies a dual-channel result. As adoption rises, the rolling out-of-sample R^2 against realised returns climbs from roughly 0.09 to 0.19 through coordinated adopter demand, while the same forecast's R^2 against the demand-adjusted return falls from roughly 0.07 to 0.04, with the rolling effective AR coefficient rising from 0.32 to 0.45 in step. Economic value does not erode in the cost-free baseline. The headline summary is therefore that adoption raises realised predictive performance through a self-fulfilment channel while reducing the independent demand-adjusted signal the forecast was originally designed to exploit.

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